

# Math 115A — Discussion Section

Session 01: Fields

Thursday, September 24, 2026

Lecture material: Lecture 1 (Friedberg §1.1–1.2)

## Learning Objectives

By the end of this section, students should be able to:

- ☐ State the field axioms and verify whether a given algebraic structure satisfies them.
- ☐ Explain why  $\mathbb{Q}$  (not  $\mathbb{Z}$  or  $\mathbb{N}$ ) is the natural home for Gaussian elimination.
- ☐ Compute addition and multiplication in  $\mathbb{F}_2$  and  $\mathbb{F}_3$  and verify the field axioms hold.
- ☐ Identify why  $\mathbb{Z}/4\mathbb{Z}$  fails to be a field by exhibiting a non-invertible nonzero element.

## Common Student Errors

- **“Division” vs. “inverse.”** Students say “you can divide” without knowing what that *means* axiomatically. Pin it down:  $a^{-1}$  exists in the field, so “ $a/b$ ” is *defined as*  $a \cdot b^{-1}$ .
- **Confusing  $\mathbb{Z}/n\mathbb{Z}$  with  $\mathbb{F}_n$ .**  $\mathbb{Z}/n\mathbb{Z}$  is always a ring; it is a field *only* when  $n$  is prime. The notation  $\mathbb{F}_p$  emphasizes it is a field.
- **$1+1=0$  in  $\mathbb{F}_2$  feels wrong.** Reassure: the elements are just labels. This is a valid algebraic structure; our usual intuitions about  $\mathbb{R}$  do not apply.
- **Forgetting to reduce mod  $p$ .** Students sometimes write  $2 \cdot 2 = 4$  in  $\mathbb{F}_3$  instead of 1.

## Plan (60 minutes)

### Intro (2 min)

Introduce yourself briefly. Tell students the goal of discussion: not to re-do lecture, but to slow down on the foundational material and actually *work* with the definitions. Today: we start at the very bottom — what is a field?

### Part 1 — The Field Axioms (10 min)

**Board:** Write the heading “What is a field?” and build up the following definition step by step.

A **field** is a triple  $(\mathbb{F}, +, \cdot)$  consisting of a set  $\mathbb{F}$  with two operations (closure is built into the operations), satisfying:

**Additive axioms:**

- (F1) Associativity:  $(a + b) + c = a + (b + c)$  for all  $a, b, c \in \mathbb{F}$
- (F2) Zero:  $\exists 0 \in \mathbb{F}$  with  $0 + a = a = a + 0$  for all  $a$
- (F3) Commutativity:  $a + b = b + a$  for all  $a, b \in \mathbb{F}$
- (F4) Opposite: for every  $a \in \mathbb{F}$ ,  $\exists b \in \mathbb{F}$  with  $a + b = 0$

**Multiplicative axioms:**

- (F5) Associativity:  $(a \cdot b) \cdot c = a \cdot (b \cdot c)$  for all  $a, b, c \in \mathbb{F}$
- (F6) One:  $\exists 1 \in \mathbb{F}$  with  $1 \cdot a = a = a \cdot 1$  for all  $a$
- (F7) Commutativity:  $a \cdot b = b \cdot a$  for all  $a, b \in \mathbb{F}$
- (F8) Distributivity:  $a \cdot (b + c) = a \cdot b + a \cdot c$  for all  $a, b, c \in \mathbb{F}$
- (F9) Non-triviality:  $0 \neq 1$
- (F10) Invertibility: for every  $a \in \mathbb{F}$  with  $a \neq 0$ ,  $\exists b \in \mathbb{F}$  with  $a \cdot b = 1$

**Board:** Box the punchline: *In a field you can add, subtract, multiply, and divide by any nonzero element, and the usual algebraic laws hold.* Keep the axiom list visible for the rest of the session.

## Part 2 — Where Do Fields Come From? The $\mathbb{N} \rightarrow \mathbb{Z} \rightarrow \mathbb{Q}$ Story (12 min)

**Board:** Draw the chain  $\mathbb{N} \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{R} \longrightarrow \mathbb{C}$

Walk through each level, asking students what fails at each step.

- $\mathbb{N} = \{0, 1, 2, \dots\}$ : No additive inverses —  $3 - 5 \notin \mathbb{N}$ . Not even a group under addition.
- $\mathbb{Z}$ : Additive inverses exist, but multiplicative inverses do not:  $2 \in \mathbb{Z}$  has no inverse in  $\mathbb{Z}$ . Not a field.
- $\mathbb{Q}$ : Every nonzero  $\frac{p}{q}$  has inverse  $\frac{q}{p}$ . All axioms hold.  **$\mathbb{Q}$  is the first field in this chain.**

### Gaussian elimination motivation.

**Board:** Write the system

$$2x + 3y = 1, \quad 4x + y = 2.$$

Row reduction requires dividing a row by a pivot. Over  $\mathbb{Z}$ , dividing by 2 is not guaranteed to stay in  $\mathbb{Z}$ . Over  $\mathbb{Q}$ , you always can. Axiom (F10) is *exactly* what makes Gauss–Jordan work.

**Punchline:** The Math 33A algorithm works because  $\mathbb{R}$  (and  $\mathbb{Q}$ ) are fields. In 115A, every theorem involving solving  $T(\mathbf{v}) = \mathbf{w}$  or finding bases implicitly uses this.

## Part 3 — What Is Special About $\mathbb{R}$ and $\mathbb{C}$ ? (8 min)

$\mathbb{Q}$  is a field, but  $\mathbb{R}$  and  $\mathbb{C}$  carry additional structure.

- $\mathbb{R}$ : The *completion* of  $\mathbb{Q}$  — filling in all gaps like  $\sqrt{2}$  and  $\pi$ . Completeness is what analysis (131A/B) is about. For 115A,  $\mathbb{R}$  is our most common base field.

- $\mathbb{C}$ : Even  $\mathbb{R}$  has a gap:  $x^2 = -1$  has no real solution. Adjoin  $i = \sqrt{-1}$  to get  $\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}$ .

The key virtue of  $\mathbb{C}$ : **every polynomial with complex coefficients has a root in  $\mathbb{C}$**  (Fundamental Theorem of Algebra). This matters when we reach eigenvalues in Chapter 5: the characteristic polynomial always splits over  $\mathbb{C}$ .

**Board:** Annotate the chain:

$$\begin{array}{ccccc} & & \mathbb{Q} & \subset & \mathbb{R} & \subset & \mathbb{C} \\ & \text{first field} & & \text{complete (analysis)} & & \text{algebraically closed} & \end{array}$$

#### Part 4 — Finite Fields: $\mathbb{F}_2$ , $\mathbb{F}_3$ , and Why $\mathbb{Z}/4$ Fails (20 min)

**Board:** New heading: “Unexpected fields —  $\mathbb{F}_p$ ”

Fix a prime  $p$ . Define  $\mathbb{F}_p = \{0, 1, \dots, p-1\}$  with addition and multiplication done modulo  $p$ .  
 $\mathbb{F}_2 = \{0, 1\}$  — **work it out together (5 min)**.

**Board:** Draw both tables side by side.

|   |   |   |
|---|---|---|
| + | 0 | 1 |
| 0 | 0 | 1 |
| 1 | 1 | 0 |

|   |   |   |
|---|---|---|
| · | 0 | 1 |
| 0 | 0 | 0 |
| 1 | 0 | 1 |

Ask students: what is the additive inverse of 1 in  $\mathbb{F}_2$ ? (Answer: 1, since  $1 + 1 = 0$ .) What is the multiplicative inverse of 1? (Answer: 1.) Is every nonzero element invertible? (Yes — the only nonzero element is 1, and  $1 \cdot 1 = 1$ . ✓)

$\mathbb{F}_3 = \{0, 1, 2\}$  — **students fill in the tables (5 min)**.

**Board:** Draw blank tables; have students fill them in.

|   |   |   |   |
|---|---|---|---|
| + | 0 | 1 | 2 |
| 0 | 0 | 1 | 2 |
| 1 | 1 | 2 | 0 |
| 2 | 2 | 0 | 1 |

|   |   |   |   |
|---|---|---|---|
| · | 0 | 1 | 2 |
| 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 2 |
| 2 | 0 | 2 | 1 |

**Board:** Circle the key entry:  $2 \cdot 2 = 4 \equiv 1 \pmod{3}$ , so  $2^{-1} = 2$  in  $\mathbb{F}_3$ .

Every nonzero element is invertible. ✓

$\mathbb{Z}/4\mathbb{Z}$  — **show this is NOT a field (7 min)**.

**Board:** Elements  $\{0, 1, 2, 3\}$ , operations mod 4. Ask: is 2 invertible?

Seek  $x$  with  $2x \equiv 1 \pmod{4}$ :

$$2 \cdot 1 = 2, \quad 2 \cdot 2 = 4 \equiv 0, \quad 2 \cdot 3 = 6 \equiv 2.$$

No element gives 1. So  $2 \in \mathbb{Z}/4\mathbb{Z}$  is nonzero but has no multiplicative inverse. **Axiom (F10) fails.**  
 $\mathbb{Z}/4\mathbb{Z}$  is **not** a field.

Why?  $4 = 2^2$  is not prime, and 2 is a *zero divisor*:  $2 \cdot 2 = 0$  even though  $2 \neq 0$ . A field can never have zero divisors (exercise: prove this from the axioms).

**Board:** Box the moral:

$$\mathbb{Z}/n\mathbb{Z} \text{ is a field} \iff n \text{ is prime.}$$

## Wrap-up (3 min)

Today's takeaway: the word "field" in "vector space over a field  $F$ " is load-bearing.  $F$  is the set of scalars. Scaling must support division so that linear algebra works.

**Next session (Session 02, Tue Sep 29):** vector spaces. Now that we know what a field is, the definition makes sense: a set of "vectors" you can add and scale by elements of a field.

## Board Layout Guide

Rough sketch for Part 4:

LEFT PANEL

RIGHT PANEL

-----  
 Field axioms (stay up all session)

-----  
 F\_2 tables | F\_3 tables  
 (add, mult) (add, mult)

Z/4: show  $2 \cdot x$  for  $x=1,2,3$   
 $\rightarrow$  none gives 1  $\rightarrow$  NOT a field

Write the chain  $\mathbb{N} \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{R} \rightarrow \mathbb{C}$  on the center board during Part 2 and leave it up through Part 3.

## Friedberg References

- **Section:** §1.1 (Fields), pp. 1–10 (5th ed.)
- **Suggested exercises (§1.1):** Exercise 1 (verify field axioms for  $\mathbb{Q}$  and  $\mathbb{C}$ ); Exercise 3 (show  $\mathbb{Z}/p\mathbb{Z}$  is a field for  $p$  prime); Exercise 4 (zero divisors and fields)